
Force sketching in MD: can a few random probes replace multi-head committee force uncertainty?

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Abstract

Committee disagreement is the standard uncertainty signal for machine-learned interatomic potentials, and a multi-head committee makes it nearly free for *energies*: M heads share one trunk, so a single forward pass returns all M . Forces are not free. Forces are gradients, so each head’s forces need their own backward pass, and force disagreement — the quantity acquisition actually consumes — costs about M times more. We ask whether $K \ll M$ cheap random probes can estimate it instead, on a pretrained eight-head MACE committee across 3BPA and three rMD17 molecules. The answer is *no* for replacement and *yes* for screening. Even given six of the seven probes the exact computation would use, the best estimator recovers only 72–80% of the genuinely most uncertain structures, and we diagnose why: the maximum over $3N$ noisy component estimates is biased upward, which no marginal finite- K correction removes. But a conformally calibrated gate built from the same cheap score skips 84–87% of the expensive computations while still catching 96–98%, beating a free energy-disagreement gate and head subsampling at matched budget in 11 of 12 comparisons. We report the wall-clock caveat plainly: the saving is $1.85\times$ in batch but only $1.11\times$ at batch size one, which is the single-trajectory MD regime.

1 Why force uncertainty is the expensive half

Uncertainty quantification is what makes a learned potential usable beyond its training distribution: it flags extrapolation during molecular dynamics and decides when a structure must go back to the quantum-chemistry calculation the model was trained to imitate. Committee disagreement remains the most widely used estimator of that error — where training data pinned the model down its members agree, and where it did not they diverge, so their spread stands in for an error we cannot see [Behler, 2015, Podryabinkin and Shapeev, 2017, Carrete et al., 2023]. Multi-head committees [Beck et al., 2025], and shallow ensembles more broadly [Kellner and Ceriotti, 2024], make this cheap by attaching M readout heads to one shared message-passing trunk, so all M energies come from a single forward pass instead of M separately trained models.

That amortization does not extend to forces. As the multi-head committee work states directly, “a single backward pass can not yield the forces for all the heads ... the standard deviation requires multiple reverse passes through the computational graph.” Forces are what MD and most acquisition rules consume, so the expensive half of committee uncertainty is exactly the half the shared trunk fails to amortize. Downstream, though, one rarely needs every member force: acquisition needs a *ranking* and a safety gate needs a *threshold decision*. That motivates estimating the disagreement statistic directly rather than reconstructing the member forces that define it.

2 Sketching the centered head subspace

For a structure of N atoms with coordinates $x \in \mathbb{R}^D$, $D = 3N$, the committee returns head energies $E(x) \in \mathbb{R}^M$ from one forward pass, and head forces $f_m = -\nabla_x E_m$ collected as columns of

$F \in \mathbb{R}^{D \times M}$. With $P = I_M - \frac{1}{M} \mathbf{1}\mathbf{1}^\top$ the centering projector, the per-coordinate disagreement and its per-structure sum are

$$v_d = \frac{1}{M-1} \sum_{m=1}^M (F_{dm} - \bar{F}_d)^2, \quad S = \sum_{d=1}^D v_d = \frac{\|FP\|_F^2}{M-1}. \quad (1)$$

Reverse mode returns the gradient of *one scalar* per backward pass; we call one such pass a *lane*. Since S depends on F only through FP , whose columns sum to zero, it lives in an $(M-1)$ -dimensional centered subspace with orthonormal basis Q : differentiating $q_j^\top E$ costs one lane and returns $g_j = Fq_j$, with $v_d = \frac{1}{M-1} \sum_j g_{j,d}^2$. Exact force uncertainty therefore costs M lanes — $M-1$ centered directions plus the mean force MD needs anyway. *Our question is whether downstream decisions need all $M-1$ of them.*

Probes. Instead of all $M-1$ directions, probe the subspace with $K \ll M-1$ random ones and rescale. Drawing $z_k \sim \mathcal{N}(0, I_M)$ and centering, $w_k = Pz_k$, one lane returns $g_k = Fw_k$ and $\hat{v}_d = \frac{1}{K(M-1)} \sum_k g_{k,d}^2$ is unbiased. Random probes can overlap and waste budget, so we instead draw a Haar-uniform orthonormal K -frame inside the centered subspace. Writing $r = M-1$, the two have relative standard deviations $\sqrt{2/K}$ and $\sqrt{2(r-K)/(K(r+2))}$: orthogonalization multiplies estimator variance by exactly $(r-K)/(r+2)$, a factor of $4/9$ at $K=3, M=8$. It costs nothing extra and is the largest single lever we find. These estimators are the molecular-UQ instance of randomized trace and norm estimation [Hutchinson, 1990, Avron and Toledo, 2011, Meyer et al., 2021]; the orthogonal variant is the head-space analogue of orthogonal random features [Yu et al., 2016].

Control variate. Disagreement is not spread evenly over the subspace, so probing it obviously wastes budget. Splitting $P = Q_{r_0}Q_{r_0}^\top + R$, we evaluate the r_0 leading directions of a design-split average exactly and sketch only the residual with $K' = K - r_0$ Haar probes; the estimator stays unbiased.

Screening instead of replacement. If a sketch cannot *replace* the exact score, it can still certify when that score is small. Split conformal [Vovk et al., 2005, Angelopoulos et al., 2024] turns any score into a calibrated bound from held-out data alone: form the ratios $\eta_i = S(x_i)/(\hat{S}(x_i) + \epsilon)$ on a calibration split and let c_α be their $\lceil (n+1)(1-\alpha) \rceil$ -th order statistic. Then $U(x) = c_\alpha \hat{S}(x)$ sits above $S(x)$ at least $1-\alpha$ of the time, so $U(x) < \tau$ *certifies* that the exact score is below the high-uncertainty threshold: skip when it is, fall back otherwise.

3 Experiments

We use the pretrained eight-head MACE [Batatia et al., 2022] committees released with Beck et al. [2025] on 3BPA [Kovács et al., 2021] at 1200 K and the joint rMD17 [Christensen and von Lilienfeld, 2020] committee on ethanol, aspirin and azobenzene. Each system is split once into disjoint design (20%, fitting the control-variate basis and the threshold τ), calibration (20%, fitting c_α alone) and test (60%) parts. Following Beck et al. [2025], structures are ranked by the maximum disagreement over force components; we take that as primary and report the global sum as an easier secondary. Estimator evaluation is in float64; timing is float32 on one RTX 5070 Laptop GPU. Ten seeds are frozen before testing and none discarded; intervals are paired bootstraps [Efron and Tibshirani, 1993] over structures with 10,000 resamples.

4 Results

4.1 Replacement fails, and we can say why

Figure 1 gives the negative result. Under the primary max-component rule the best sketch reaches only 0.484–0.570 top-5% recall at $K=4$ and 0.724–0.801 even at $K=6$ of 7 — the 0.90 line is not attained at any budget short of the exact basis.

The mechanism is specific to extreme-value acquisition. Making each $\hat{\sigma}_d$ unbiased *marginally* does not make $\max_d \hat{\sigma}_d$ unbiased: a maximum over $3N$ noisy estimates is biased upward, and the bias grows with the estimator’s noise. Measured directly, the score is inflated by 1.42–1.53 at $K=1$,

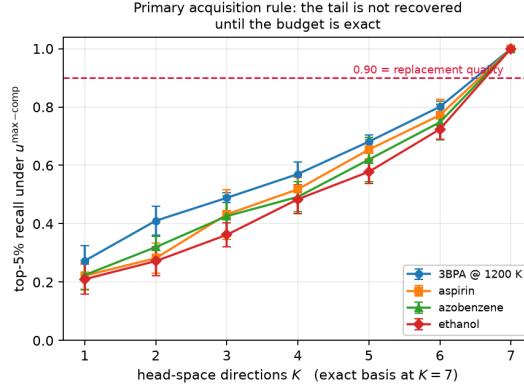


Figure 1: Top-5% recall against probe budget K under the primary max-component rule, on all four systems. Recall climbs steadily but never reaches the 0.90 replacement line short of the exact basis at $K=7$. Error bars are the standard deviation over ten frozen seeds.

falling to 1.07–1.08 at $K=4$ and to 1.000 at the full basis where the estimator is exact — a useful check on the pipeline. This obstructs replacement but not screening: the bias is monotone in the estimator’s noise, so a scale-calibrated gate absorbs it into c_α .

4.2 Screening works, and beats the cheaper gates

A calibrated gate is a generic wrapper — any score correlated with exact force disagreement can be calibrated the same way — so the claim only stands if this gate beats the cheaper ones a reader will reach for. We build three at matched lane budget with identical splits, calibration and threshold: a *free* energy-disagreement gate (the M head energies come from the forward pass already run, costing zero uncertainty lanes); head subsampling using the *exact* mean force the sketch already pays a lane for; and a plain Haar sketch.

gate	lanes	exact skipped	high-UQ recall	speedup
energy disagreement (free)	1	0.294–0.525	0.636–1.000	1.05–1.33×
head subsampling, exact mean	5	0.474–0.747	0.847–0.936	1.16–1.33×
Haar sketch	5	0.525–0.734	0.870–0.926	1.19–1.32×
control variate	5	0.836–0.872	0.964–0.982	1.40–1.43×

Ranges across the four systems, $r_0=2$, $K=4$, $\alpha=0.05$, on the held-out test split. Every gate pays one mean-force lane; on fallback each completes the exact basis with $r-K$ further passes rather than a fresh M , charged identically for all of them.

The control variate *Pareto-dominates* — strictly more skipped *and* strictly higher recall — in 11 of the 12 gate-by-system comparisons, including every one against the two sketch baselines. The exception is worth stating: on ethanol the free energy gate attains perfect recall (1.000) while skipping only 0.322. Elsewhere that free gate is the informative failure, collapsing to 0.636 recall on azobenzene — energy disagreement is a genuinely different signal from force disagreement, not a cheap proxy for it.

4.3 The wall-clock caveat, stated plainly

Against a *batched* exact baseline the total force-plus-UQ speedup at $K=3$ is $1.85\times$ at $B=16$, but collapses to $1.11\times$ at $B=1$, where batched reverse mode makes extra cotangent lanes nearly free and the gate is a net slowdown ($0.93\times$). A single MD trajectory advances one structure at a time, so $B=1$ is the trajectory regime: this method pays off for batched active-learning sweeps over candidate structures, not for stepping one trajectory. Making that baseline available at all required disabling the TensorExpr fuser, without which batched reverse mode fails on this stack; had we not, we would have compared against a Python loop and overstated every speedup here.

5 Conclusion

Exact multi-head force uncertainty cannot be replaced by a handful of randomized probes at the fidelity acquisition requires, and the obstruction is an extreme-value bias no marginal correction removes. It can be *screened*: a conformally calibrated gate over a control-variate sketch avoids 0.836–0.872 of exact computations while retaining 0.964–0.982 of high-uncertainty structures on every system. That is a property of the estimator; whether it saves time is a separate question whose answer is regime-dependent. We preserve a proxy, not a demonstrated outcome: no active-learning loop is run and no reference-force error measured, so we claim no gain in downstream model quality. Committee disagreement is only a proxy for error; calibration is marginal and under exchangeability; $M=8$ bounds the maximum saving; and rMD17 spans only 9–24 atoms.

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